

ACHINSK, Russian Federation

WMO: 294670

Lat: 56.283N

Lon: 90.517E

Elev: 257

StdP: 98.27

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-30.8	-37.0	0.1	-34.2	-33.5	0.2	-30.7	9.9	-7.5	9.1	-9.0	1.6	230	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	29.1	18.7	27.2	18.0	25.4	17.3	20.1	26.4	19.2	25.1	18.4	23.8	2.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.3	22.9	17.0	12.5	21.5	16.2	11.9	21.3	59.1	26.4	56.0	25.1	53.3	23.9	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.5	7.4	6.4		DB	-35.4	31.8	5.1	1.8	-39.1	33.1	-42.1	34.1	-45.0	35.2	-48.7	36.5
				WB	-35.1	21.4	4.9	1.2	-38.6	22.3	-41.4	23.0	-44.1	23.7	-47.6	24.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-15.0	-11.2	-4.9	3.1	10.6	17.6	18.9	16.6	9.2	1.6	-8.5	-12.7	(6)
	DBStd	13.39	8.24	6.88	6.35	5.16	5.90	4.41	3.36	3.76	4.73	5.47	8.10	8.00	(7)
	HDD10.0	3714	777	594	462	215	67	4	0	2	72	264	554	703	(8)
	HDD18.3	6026	1035	827	721	458	248	64	34	79	276	518	804	962	(9)
	CDD10.0	855	0	0	0	7	86	231	276	205	47	4	0	0	(10)
	CDD18.3	126	0	0	0	0	9	41	51	24	1	0	0	0	(11)
	CDH23.3	928	0	0	0	1	72	336	341	170	8	0	0	0	(12)
	CDH26.7	213	0	0	0	0	15	84	80	33	1	0	0	0	(13)
(14) Wind	WSAvg	2.9	3.2	3.2	3.3	3.4	3.0	2.1	1.9	2.0	2.5	3.2	3.5	3.4	(14)
(15) Precipitation	PrecAvg	422	14	12	12	24	38	57	75	67	41	33	26	23	(15)
	PrecMax	631	35	33	33	52	106	127	172	135	75	87	50	53	(16)
	PrecMin	274	3	2	2	3	6	16	7	24	13	9	6	4	(17)
	PrecStd	90	8	7	6	12	22	31	42	31	18	18	10	11	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.4	8.7	19.7	29.0	31.6	31.1	29.7	24.3	17.2	9.5	1.9	(19)
		MCWB	-0.1	-0.1	3.5	9.8	15.7	18.7	20.6	19.9	15.2	9.8	6.3	-0.3	(20)
	2%	DB	-0.9	0.5	6.5	16.2	25.3	29.0	28.8	27.3	21.2	13.8	5.7	0.2	(21)
		MCWB	-2.6	-1.6	2.2	8.6	13.9	18.3	19.4	18.9	14.2	8.6	3.3	-1.4	(22)
	5%	DB	-2.6	-0.9	4.9	13.7	22.6	26.9	26.8	25.1	18.7	11.5	3.5	-1.5	(23)
		MCWB	-3.9	-2.6	1.4	7.1	12.8	17.5	18.6	17.7	13.0	7.2	1.2	-2.8	(24)
	10%	DB	-4.4	-2.3	3.1	11.1	20.3	25.0	25.0	23.2	16.3	9.2	1.5	-3.3	(25)
		MCWB	-5.8	-3.9	0.2	5.7	11.8	16.9	17.9	16.8	11.8	6.0	-0.2	-4.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	0.3	4.3	11.1	16.4	20.6	21.6	21.1	16.0	10.7	6.5	0.1	(27)
		MCDB	1.4	2.2	7.4	18.4	26.1	27.7	28.3	27.8	22.1	14.8	9.2	1.5	(28)
	2%	WB	-2.5	-1.3	2.9	9.3	14.8	19.4	20.4	19.7	14.7	9.3	3.3	-1.3	(29)
		MCDB	-1.2	0.0	6.0	15.1	22.3	26.3	26.7	25.5	20.0	13.1	5.3	0.2	(30)
	5%	WB	-3.8	-2.5	1.5	7.6	13.7	18.6	19.5	18.5	13.6	7.8	1.4	-2.7	(31)
		MCDB	-2.5	-1.0	4.4	12.9	20.7	25.2	25.1	23.8	17.9	11.0	3.2	-1.4	(32)
	10%	WB	-5.8	-3.9	0.3	5.9	12.5	17.7	18.7	17.5	12.3	6.2	-0.1	-4.4	(33)
		MCDB	-4.5	-2.3	2.9	10.8	19.4	23.6	23.7	22.2	15.6	9.2	1.3	-3.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	7.0	7.8	9.2	9.2	8.4	8.3	7.2	5.7	5.5	5.5	(35)
		MCDBR	6.3	6.0	7.9	11.5	11.9	12.2	11.3	11.3	11.5	8.6	5.9	5.4	(36)
		MCWBR	5.4	5.1	5.5	6.5	5.6	5.1	4.4	4.9	5.7	5.1	4.0	4.7	(37)
	5% WB	MCDBR	6.1	5.9	7.2	10.8	10.6	10.7	9.6	9.7	10.1	7.8	5.6	5.4	(38)
		MCWBR	5.6	5.3	5.4	6.6	5.7	4.8	4.2	4.6	5.4	4.9	4.3	4.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.284	0.317	0.364	0.394	0.407	0.405	0.427	0.408	0.371	0.350	0.314	0.285	(40)
	taud		1.996	1.980	1.924	2.069	2.201	2.293	2.236	2.265	2.318	2.214	2.019	1.943	(41)
	Ebn at Noon		566	703	765	813	830	838	807	797	774	674	513	442	(42)
	Edh at Noon		66	101	139	141	131	122	126	115	95	81	65	54	(43)
(44) All-Sky Solar Radiation	RadAvg		0.76	1.72	3.04	4.28	4.94	5.92	5.33	4.22	2.73	1.39	0.69	0.45	(44)
	RadStd		0.09	0.15	0.25	0.30	0.54	0.52	0.34	0.36	0.35	0.13	0.10	0.06	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page